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Understanding the mechanisms of electronic ink operation

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Abstract
Owing to the unique features of electronic ink displays, including the bistability, paper-like appearance, and sunlight visibility, electronic ink displays have been applied in many Internet of Things (IoT) fields. We reviewed mechanisms that have been proposed to be essential for electro-optical behavior of electronic ink displays. This review might facilitate beginners to start their research in electronic ink studies.

KEYWORDS
bistable displays, electrophoretic display, E-paper, microcapsules

1 | INTRODUCTION

Electrophoretic displays (EPDs) have been widely used in the e-readers, electric shelf labels, smart bus stop signages, and so forth. Recently, EPDs have been further adopted to the Internet of Things (IoT) devices.[1–3] In the field of IoT and artificial intelligence (AI), people need a massive quantity of terminal sensors to collect data (also known as Big Data), which is crucial to cultivate the AI's central brain. If each sensor has at least one display to communicate with the users, then the tremendous number of sensors would consume so much electricity and workforce to replace the batteries. Because of the inherent advantage of low power consumption, EPD is regarded as one of the most promising display technologies for future IoT applications.

The superior features of EPDs result from its display media, the electrophoretic dispersion, also known as electronic ink. EPD is a kind of display technology whose operating principle is different from the conventional liquid-crystal displays (LCD) and organic light-emitting diode (OLED).[4,5] As for power consumption, compared with OLED and LCD, EPD has superior advantages owing to its inherent behavior of bistability,[6] as shown in Table 1. EPDs also have disadvantages like indoor color performance, especially under dim light ambience.[8] Electronic ink is generally made of electrophoretic particles with positive charges and negative charges both in nonpolar solution. The particles with opposite charges can be separated under the electric field and show different grayscales. However, the charging process, driving scheme, and bistability mechanisms have not been systematically established yet. Therefore, we tried to review several essential mechanisms for EPD research beginners and pointed out several challenges in this field.

2 | GENERAL VIEW OF ELECTRONIC INK OPERATION

The electronic ink that we used contained pigment particles, charge controlling agents (CCAs), nonpolar solvent, and various additives. The particles were usually surface treated and charged by CCA; thus, the particles could be

TABLE 1 Power consumption of different display technologies[7]

	Outdoor	Color (indoor)	Power consumption
PMOLED	+	+++	+
AMOLED	+	+++	++
MIP (reflective)	++	+	+++
TF-LCD (transflective)	++	++	++
LCD (transmissive)	+	++	++
E-paper	+++	+	+++

Abbreviations: AMOLED, active-matrix organic light-emitting diode; LCD, liquid-crystal display; MIP, memory in pixel; PMOLED, passive matrix organic light-emitting diode; TF-LCD, transflective liquid-crystal display.

driven by an external electric field. Therefore, the specific distribution of pigment particles could be controlled by a designed waveform. This particle distribution resulted in different reflectance, which was termed “grayscale” or “gray level.”

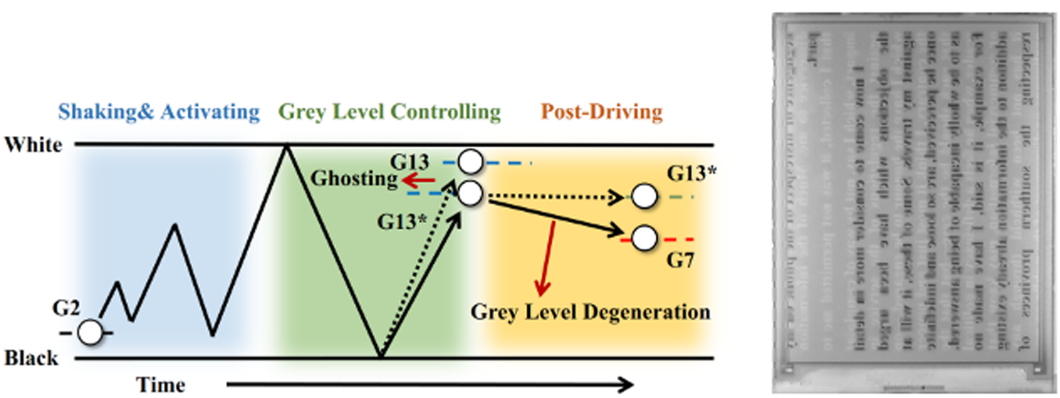
To explain the particles' behavior under the different moving stages, the optical trace of these particles during the driving process was shown in Figure 1A. In a typical driving waveform, particles were shaken by an alternating voltage to activate the particles and then driven to the upper and bottom boundaries to erase the image history. If we intended to stimulate the particles to the 13th grayscale (G13), the reflectance was eventually G13*, which meant the controlling of grayscales in the EPD was not perfect enough. This slight deviation between the target grayscale and the actual grayscale resulted in image residue, which was termed “ghosting,”[9] as shown in Figure 1B. The ghosting was usually related to the previous grayscale of the EPD. The commonly used solutions to solve the “ghosting” are multiflash transition, weakly indirect transition, local direct current (DC) balance, and global DC balance. With the multiflash shift, electrophoretic particles were pushed several times to the capsule boundaries to erase the historical grayscale. However, it introduced an annoying flash; thus, weakly indirect transition[10] was adopted to suppress the flash.

To reduce grayscale errors, the driving waveform was usually composed of history erasing phase, activation phase, and display phase.[5] Furthermore, in the process of the cyclic display, the electrophoretic particles were pushed in the same direction several times, which led to the accumulation of charge in a specific direction with the built-in field, affected the accuracy of the grayscale, and reduced the lifetime of the EPD. This charge accumulation-induced phenomenon was termed “DC non-balance.”

To fix this problem, the summation of definite pulse length of pushing the particle in a particular direction should be the same as the negative pulse length of pushing the particle in the opposite direction, so that the energy could be balanced out, which is called “DC balance.” The approach to achieve DC balance in a single driving cycle is called “Local DC Balance.”[11] To shorten the image updating time and flashing times, “Global DC Balance” approach[12] has been further adopted.

After removing the external driving voltage, the distribution of particles was not supposed to change due to the bistability characteristics of the EPD. However, the distribution of particles was still influenced by some other factors, such as the built-in electric field of opposite particles, which caused the variation of the grayscale of EPD and was usually called degeneration of grayscale.

The multi-grayscales of the EPD can be achieved by controlling the polarity and the amplitude of the voltage, which is called pulse amplitude modulation (PAM). Also, it can be obtained by controlling the polarity and the



(A) (B)

FIGURE 1 The ghosting phenomenon. (A) Schematic optical trace of particles manipulated by driving waveform and (B) the photograph of ghosting image

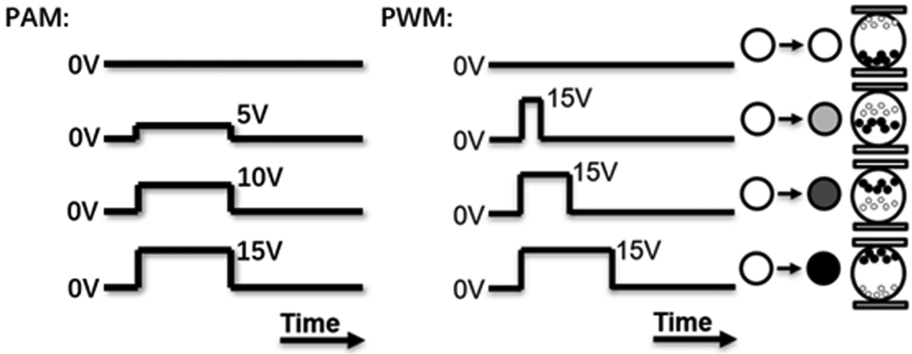


FIGURE 2 The multi-grayscale of E-paper driven by PAM and PWM

duration of the voltage applied between the two electrodes of the EPD, which is called pulse width modulation (PWM). As shown in Figure 2, researchers have utilized PWM to control the movement of electrophoretic particles, so that the spatial distribution of electrophoretic particles would be different. The grayscale was determined by the spatial distribution of the particles.[13–15]

The driving waveform could not be simply determined by the device equation,[16] because the EPD's response curves have been found to be nonlinear.[17,18] Even with the same driving waveform, different batches of electrophoretic particles may show various performances. The slight difference of particle size distribution, zeta potential, additive amounts, dielectrics' condition, and so on so forth would significantly influence the electronic ink performances. Moreover, even the same batch of electronic ink device with different driving history may also accumulate different inner energy within the electronic ink dispersion system and thus causes different particles' behavior. Therefore, a more tolerating waveform that can be applied for various conditions to show the same performance is very critical. In this regard, summarizing the particles' behaviors with a lookup table (LUT) is an economical way to simplify all these complexities for realizing electronic ink display mass production.[19] Also, the temperature is an essential factor affecting electronic ink's behavior, which directly influences the performance of the EPD.[19]

For commercial EPD, there are usually some methods of a driving system for temperature compensation.[19–24] Designing several driving waveforms for different ambient temperature have been a facile method for EPD.[19] Besides, a real-time adaptive image processing system based on the ambient temperature has been described by Kao and his co-authors, which would be another way for the temperature compensation of EPD.[21,22]

However, manual work for recognizing the "ghosting" image still costs a significant amount of time and workforce. It is still a challenge to develop an automatic system for solving this problem.

3 | CHARGING MECHANISM

The electrophoretic particles in EPD applications were prepared in nonpolar solvent. Differing from the conventional charging mechanism in polar solvents, EPDs need reversed micelles (RMs) to transport the charges between species in the solvent media.[25] In the nonpolar solvent, amphiphilic CCAs or surfactants would form RMs after reaching the critical micelle concentration (CMC).[26–28]

When the CCAs were added into electrophoretic dispersion, the hydrophobic side of the CCA dissolves and disperses in nonpolar solvents, and the hydrophilic side was adsorbed to the particles. CCAs provided dispersity and suspension to particles while CCAs assemble in RMs. In this process, some CCAs endowed the particles and RMs with charges.[29,30] The charges of the electrophoretic particles depended on the modification of particles and functional groups of CCAs.

According to the theory of electric double layer (EDL), the charges carried by RMs forms counter-ion cloud as shown in Figure 3. When an external electric field was applied, the separation of electronic ink and RMs occurred.[25] If the external electric field were healthier enough to overcome the attractive force of the dipole, the RMs would separate from the original counter-ion cloud, which increased the net charge of the particle. In this regard, the internal electric field was caused by the separation of oppositely charged particle and RMs in the EDL, which was the electrical polarization of the EDL. However, the energy was too small to be defined as the threshold energy.

In the nonpolar solvent system, the charge takes the RMs as the carrier. Some particles modified by quaternization[31] or polymerization[32] of functional monomers were charged by desorption of ions to RMs physically,[33,34] as shown in Figure 4. Lewis acid-base reaction plays a dominant role in the process of particles obtaining charge.[35] Particles were charged positively or negatively electrified, while RMs were charged oppositely. RMs donated or accepted charges in this behavior shown in Figure 4.

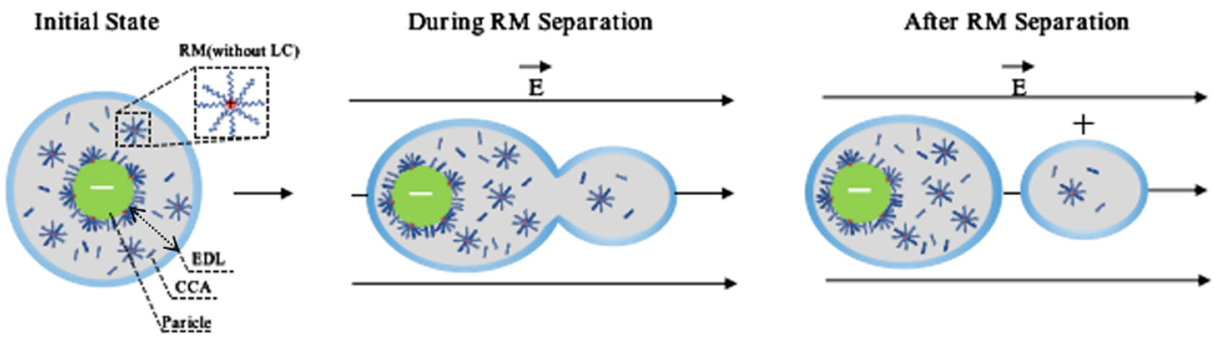


FIGURE 3 The separation of electronic ink and reversed micelles under the external electric field

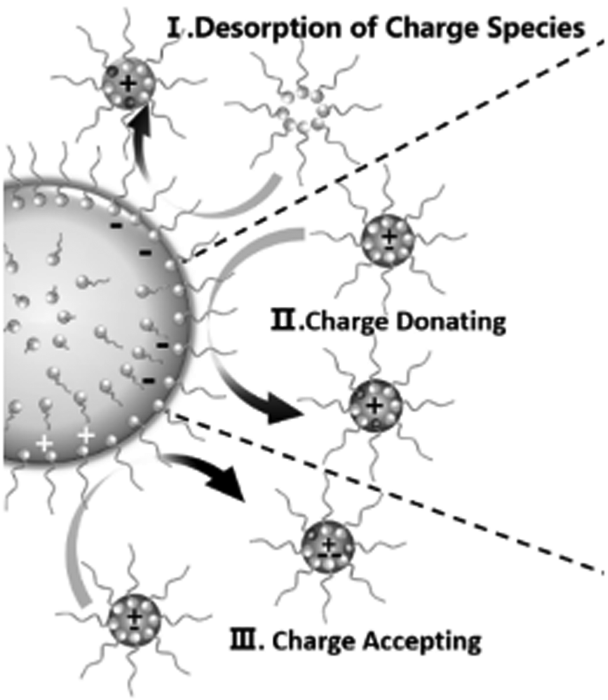


FIGURE 4 Charge exchanging between particles and RMs

4 | RHEOLOGICAL EFFECTS DURING DRIVING

The rheology of the electronic ink particles within the micro-chamber is complicated. It involves many factors and forces, such as the electrical voltage, charging amount on particles, the size of particles, the solvent's viscosity in the system of an electrophoretic dispersion, and so on. Felix Carrique et al. have mentioned that the mobility linearly connected to the charging number of particles when the charges were below a critical point and that the mobility would reach a plateau if the charges were higher than the critical point.[36] Morrison and Tarnawskyj have proposed an equation connecting the electrophoretic mobility, the charge per unit mass, the size of particles in low conductivity dispersion, and the fluid viscosity.[37] The equation illustrated that the mobility was inversely proportional to the solvent's viscosity.

To further investigate the relationship between the particles' velocity and the solvent's viscosity, iso-alkane aliphatic hydrocarbons were used. In Figure 5, the solvents A–D had different viscosities from low to high, respectively. Furthermore, an optical-electric setup for the reflective test was designed, which could gradually increase the voltage applied to the EPDs to study the optical response under varied external fields. It was suggested that the lower the viscosity of the solvent, the agiler the optical response of EPD, as shown in Figure 5, which meant that the particles encountered less resistance than particles in a more viscous solvent when moving under the applied voltage.[38,39] This finding was significant to realize full-color ink. As the full-color electronic ink accommodated several kinds of color particles to show specific color with microscale dithering, the particles would be too viscous to be driven. Thus, lowering down the viscosity is very critical.

Reducing the viscosity was an effective method to increase electrophoretic mobility, but it would deteriorate EPD's bistability in the meantime. A more feasible strategy was to realize the viscosity reduction of the suspending liquid only while applying voltage and viscosity recovering after removing voltage, which was known as inversed electrorheological (IER) effect.

Our recent progress further suggested utilizing the liquid crystal molecules' flow effect to induce the charging activation and also optimized the IER effect with better switching speed. The mobility enhancement of electronic ink would be increased by a factor of 2.8 compared with the pristine ink.[40]

The fluid behavior also varied with temperatures. As shown in Figure 6, owing to the high viscosity of the solvent, the optical response under 5°C was smaller than that of 25°C.[12,41] While further increasing the temperature up to 60°C, the optical response dropped again, which was believed to result from the desorption of RMs

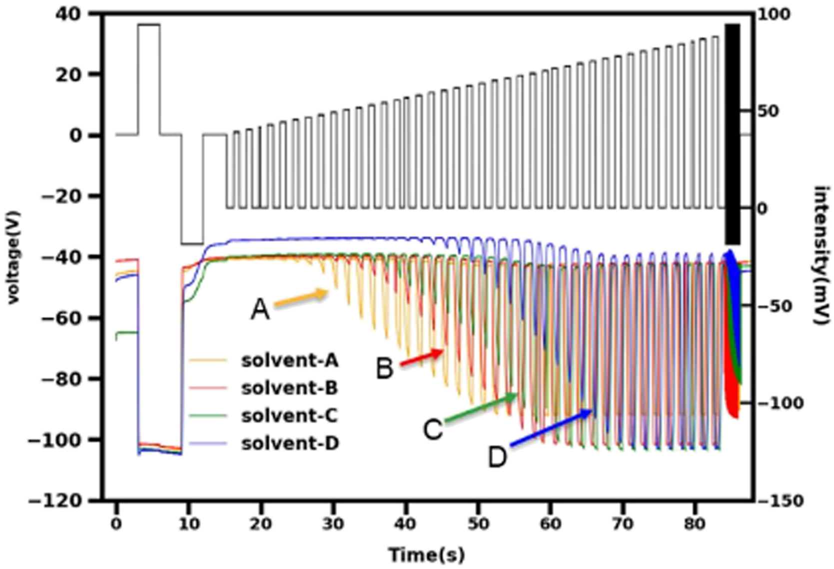


FIGURE 5 Optical response of EPDs with different solvent viscosity

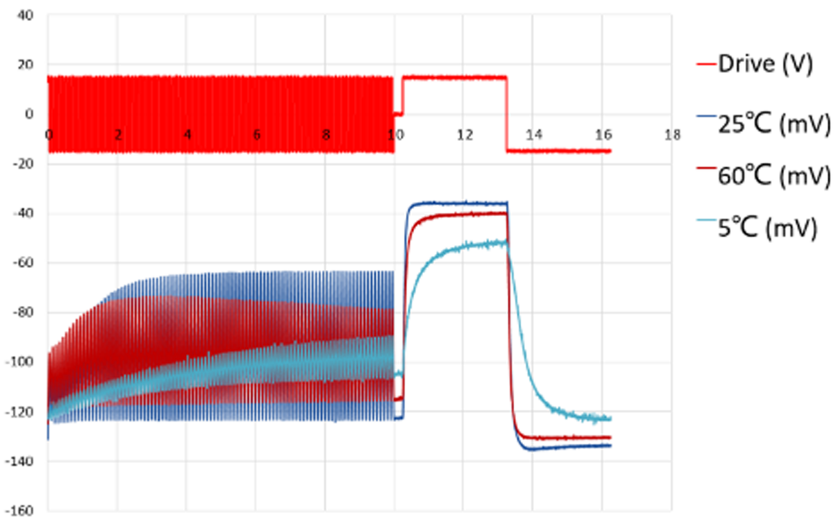


FIGURE 6 Activation of particles with various temperatures in EPD

or the less charging exchanging between the particles and CCAs under the effect of high-temperature Brownian motion.

5 | BISTABILITY AFTER REMOVING EXTERNAL FIELDS

After removing the external fields, the particles tend to stay still under the balance of forces. As shown in Figure 7, particles are acted by many forces, such as the Buoyancy, Vander Waal force, Coulombic force, frictional force, gravity, and solubility force. The capability to sustain this balance is called "bistability." This force balance is the reason why particles can stay at the same place after removing the fields and the reason why EPDs have low power consumption. To further increase the bistability, one feasible way is to utilize the depletion flocculation effect. This effect was first predicted theoretically by Oosawa et al., proposing that the colloidal particles in solution system would intend to be stable by flocculation, which was caused by the Vander Waal force and electrostatic force.[42] The depletion flocculation effect was later observed by other researchers.[43,44] Furthermore, some researchers have found that this effect could be enhanced by adding polymers and nanoparticles to the solution.[20,45,46]

Because the electronic ink was a colloidal particle system, the depletion flocculation effect could be utilized to optimize the bistability of EPD. By adding some nanoparticles or free polymers to the solvent, the pigment particles would be more stable, while still could be separated upon external field applied,[47,48] as shown in Figure 8.

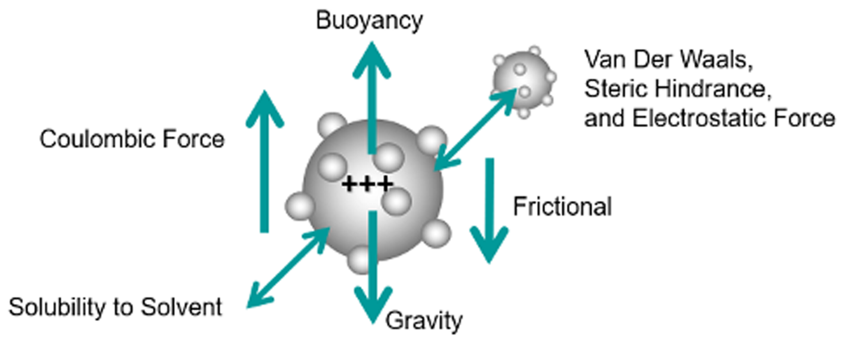


FIGURE 7 Forces encountered by electronic ink particles

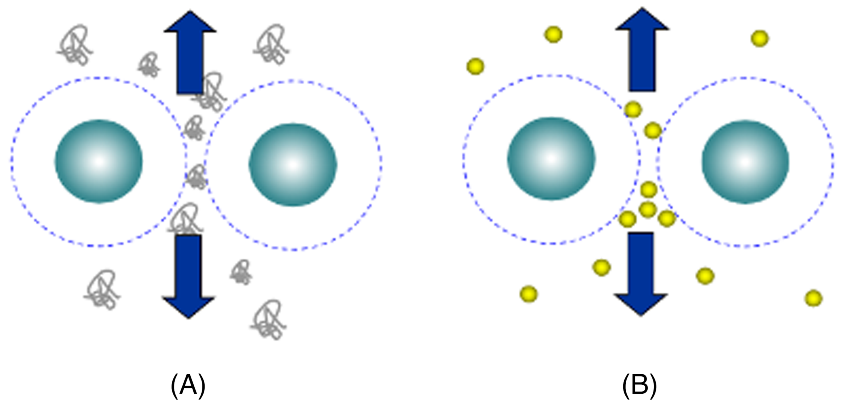


FIGURE 8 Depletion flocculation for bistability: (A) With free polymers and (B) With nanoparticles

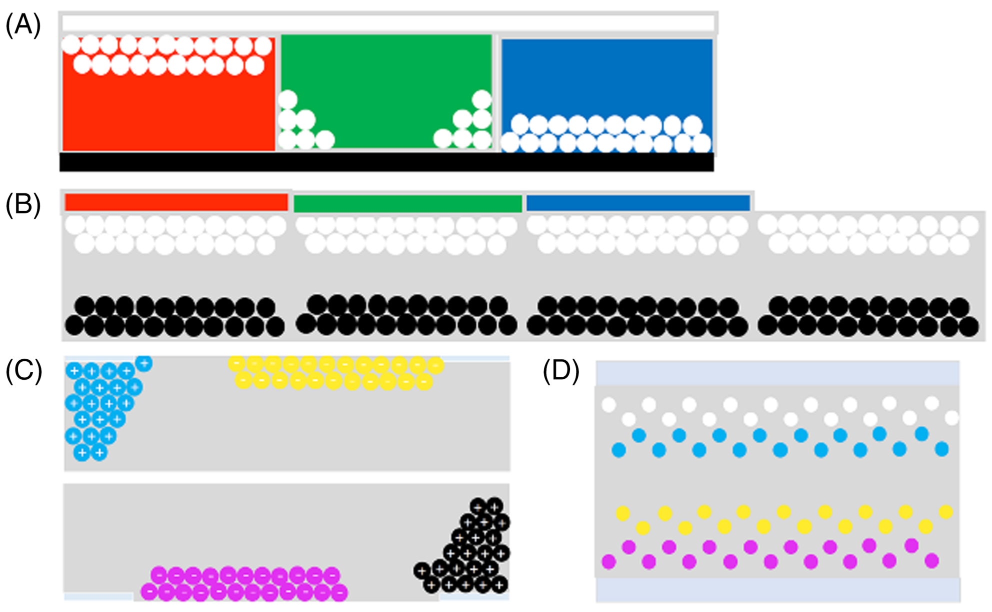


FIGURE 9 Different structures for full-color EPDs

6 | FULL-COLOR E-PAPER

Five kinds of electrophoretic color EPD are summarized below. SiPix Imaging Incorporation (later on, acquired by E ink Corporation) has provided the color states microcup-based EPD by using Dual-Mode TFT backplane, as shown in Figure 9A.[49] Colorful EPD can be achieved by color filter array, as shown in Figure 9B.[50] Figure 10C shows a colorful EPD based on in-plane electrophoresis to control colored particles.[51] A good approach to realize EPD with the multiparticle system has been demonstrated, as shown in Figure 9D.[52]

7 | CONCLUSION

In this paper, we reviewed the operational principles and fundamental mechanisms about the electrophoretic electronic ink display, including the EPD driving scheme and materials design. However, EPDs still have several challenges to be overcome, such as high switching speed with good bistability, high color performances without waveform flickering, handwriting capability with more intuitive experiences, and environmental endurances for IoT applications. The E-paper displays will be more prevailing after resolving all these challenges.

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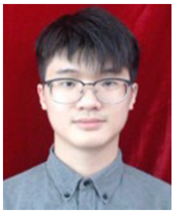

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